

Redundant Monitoring System for Contact and Proximity Detection in Human–Robot Interaction

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Abstract— Reliable detection of physical contact and spatial proximity is fundamental for safe and intuitive human-robot interaction (HRI), yet single sensing modalities often suffer from noise and occlusions. This work proposes a redundant multi-sensor monitoring framework that integrates tactile, distance, and vision-based sensing to improve the detection of physical contact events and proxemic zones in social robots. Individual sensing modalities and their corresponding redundant fusion systems were evaluated in controlled HRI experiments with 5 participants, involving touch and distance-based interactions across predefined proxemic zones. The results show that the redundant strategy achieved an accuracy of 0.96 for proximity detection and recall of 1.00 for physical contact detection. For future real-world sensing with the social robot, the redundant system can help to reduce missed detections and improve robustness for safety-oriented HRI monitoring.

I. INTRODUCTION

Social robots are increasingly being deployed in environments where they interact closely with humans, such as in healthcare, education, and homes [1]. Their effectiveness depends not only on their functional capabilities but also on their ability to perceive, interpret, and respond appropriately to human behaviors [2]. As these systems move toward more intuitive interactions, understanding human presence, movement, and contact, the sensing becomes essential for ensuring safe and engaging human-robot interactions (HRIs).

In social contexts, HRI often involves two fundamental behaviors: touch as physical human-robot interaction (pHRI) [3] and spatial approximation [4]. Touch can convey reassurance, attention, or intention, while proximity reflects social distance norms and the user's level of engagement [5]. For robots operating in shared spaces, detecting when a person approaches or establishes physical contact is crucial for adapting behaviors, maintaining safety boundaries, and responding with socially appropriate actions [2]. Consequently, accurate sensing of both contact and proximity is a key component in enabling meaningful social HRI.

Achieving reliable detection of touch and proximity requires sensing systems that are sensitive and robust under a wide range of conditions. Single sensing modalities—such as tactile sensors, distance sensors, or computer vision, often face limitations, including noise, occlusions, variable lighting, or mechanical constraints [6]. These limitations can lead to missed detections or false triggers, undermining the robot's ability to interpret human behavior consistently. A hybrid or redundant sensing approach can mitigate these challenges by combining complementary signals, increasing accuracy, and improving reliability in dynamic social environments [7].

Among available modalities, vision-based sensing offers a rich and flexible source of information for monitoring HRI [6]. Video enables the detection of physical contact, body movements, and approach trajectories [8]. However, vision alone is often insufficient due to occlusions and environmental variability, making it more suitable as part of a multi-sensor framework or for use in controlled experiments.

This work presents a redundant multimodal monitoring system that combines tactile, distance, and vision-based sensing to improve physical contact and proximity detection in HRI, while maintaining the low-cost constraints of the social robot used. The system is evaluated under controlled interaction and approach scenarios by comparing individual sensing modalities with their fused configurations, using video-based ground truth to quantify detection accuracy, timing, and error. The main contribution, is the evaluation of low-cost sensing modalities and their fusion, assessing if redundancy enhances robustness and reliability without increasing system complexity or cost. The proposed evaluation framework is illustrated in Figure 1.

II. RELATED WORK

A. Vision-based sensors

Vision-based approaches have been widely used both for direct tactile sensing (camera-in-skin / vision-based tactile sensors) and for external observation of interaction events. Embedded vision tactile sensors can estimate contact position, contact area, and force distribution by mapping images of deforma-

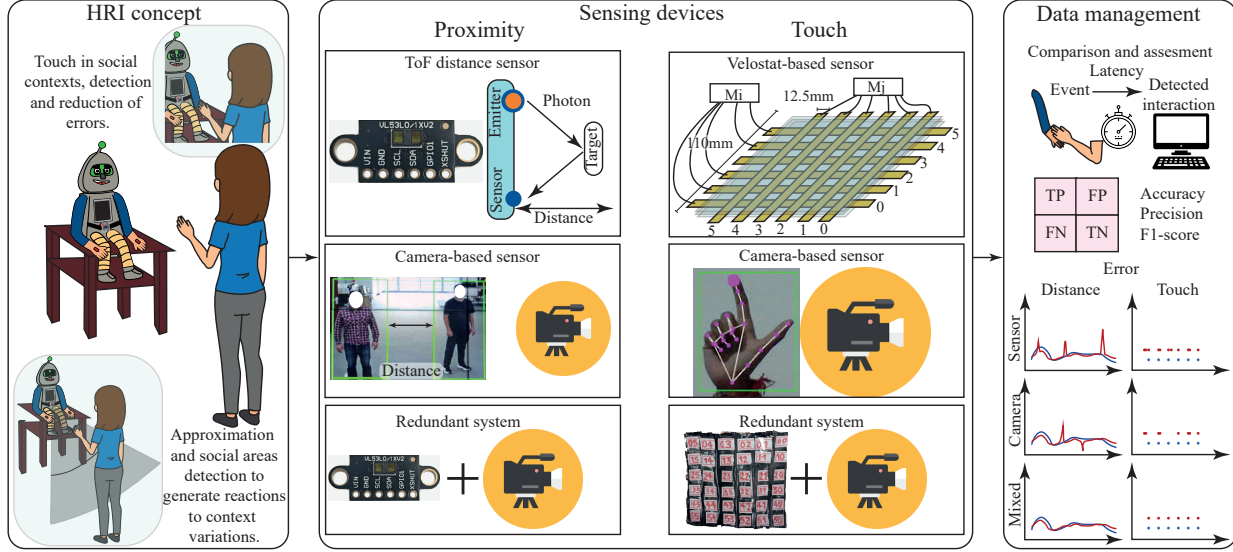


Fig. 1: Framework for the contribution stages of this work to identify the best-suited sensor for pHRI with a social robot's hand.

tion to tactile variables using learning methods [9]. Reported pipelines show good localization of contact and feasible force estimates after training, demonstrating the vision's high spatial resolution for contact inference. Externally-mounted cameras are commonly used to detect approach trajectories and to infer entry/exit of social-distance zones in different applications [10]. Vision algorithms range from bounding-box, skeleton-based estimators, to more advanced pose- and depth-based methods [8], [9], [10]. However, occlusions, lighting changes, and depth accuracy at close ranges are acknowledged limitations when cameras are used alone for HRI detection.

B. Redundant sensing

Redundancy has been proposed and implemented to increase the robustness of contact and proximity detection. A practical example is a combined time-of-flight (ToF) and self-capacitance sensor array developed for human-cooperative robots [11]. Tsuji *et al.* developed a prototype integrating 24 modules around a robot arm and could detect objects in the 300-0 mm range and register touch events [12]. Other works have proposed fusion frameworks for redundant sensing (simple logical rules, e.g., OR/AND, weighted voting, and temporal filtering). Surveys of sensor-based control and proximity perception discuss how redundancy reduces false positives and enables reactive behaviors by exploiting the complementary strengths of modalities [4]. Empirical studies commonly show that fused systems reduce missed detections and improve reliability compared to single-modality systems, though exact quantitative gains depend on sensor quality, fusion method, and application scenario.

C. Measurements in Social Robotics

Proxemics and contact behaviors have been widely studied in HRI to characterize how humans approach and interact with robots [5]. Studies show that interpersonal distance follows social norms similar to human-human interactions, influenced by robot appearance and context, and define standard proxemic zones (intimate, social, public) [13]. In addition, pHRI research highlights the importance of accurately detecting contact types (e.g., tap, sustained touch) for appropriate responses [14], [15].

Vision-based approaches provide rich spatial information but are sensitive to occlusions, illumination changes, and computational cost. In contrast, distance and tactile sensors offer low-latency and direct measurements but have limited coverage and may suffer from noise. Redundant sensing frameworks combine these modalities to improve robustness and reduce missed detections, motivating the low-cost multimodal strategy adopted in this work.

III. METHODOLOGY

This section describes the sensing systems for both measurements, the procedure maintained to compare the single and fusion detection, and the data management of errors and latency for the analysis.

A. Sensor systems

Distance sensor: The VL53L3CX is a compact Time-of-Flight (ToF) distance sensor that estimates range by measuring the time elapsed between the emission of an infrared laser pulse

and the detection of its reflection by a single photon avalanche diode (SPAD) array [16]. Based on the measured distance, three proxemic zones commonly used in human–robot interaction were defined: intimate (≤ 45 cm), social (45–120 cm), and public (120–365 cm) [13].

Touch sensor: The Velostat-based sensor was constructed as a 110×100 mm rectangular structure composed of four layers, forming a 6×6 matrix with 36 sensing points. Copper tape traces arranged vertically and horizontally defined the columns and rows, with a Velostat sheet (3M, USA) placed between them to enable pressure sensing. The assembly was insulated with tape and mounted in the hand of the robot, providing an on/off detection.

Camera-based proximity detection: The camera-based proximity detection estimates the human–robot distance from RGB video recorded by a fixed camera covering all proxemic zones. Although interactions occur in real time, the data are processed offline. Human detection is performed frame by frame using a YOLOv8 model, which provides a bounding box around the participant.

The horizontal center of the bounding box is mapped to real-world distance through a linear pixel-to-meter calibration. The participant–robot distance is computed relative to the known camera position and classified into intimate, social, or public zones. Consecutive frames within the same zone are grouped into continuous proximity intervals.

Camera-based touch detection: The camera-based detection identifies physical contact events using visual information extracted from RGB video recordings processed offline after the experimental sessions. Object detection models are applied frame by frame to detect the robot and the participant’s hand.

A touch event is inferred when spatial interaction between the detected bounding boxes is observed, based on overlap measures and the relative position of the hand within the robot’s bounding box. Temporal filtering is employed to improve robustness by requiring contact persistence over consecutive frames and applying a cooldown period to prevent duplicate detections.

Redundant sensing system: For proximity detection, the initial distance is estimated using the camera, and subsequent zone transitions are compared with those identified by the ToF sensor. If both sensors agree, the transition is accepted. In case of disagreement, the variation between consecutive measurements is computed for each sensor, and the one with the smallest variation (minimum delta) is selected as the most reliable, defining the final proximity zone.

For touch detection, a logical OR fusion strategy is adopted, where contact is detected if at least one sensing modality identifies a touch event.

B. Procedure

Figure 2 summarizes the experimental setup and procedure adopted with 5 participants (1 female and 4 males). The figure also illustrates the social robot instrumentation, including the integration and placement of the proximity and touch sensors used in the protocol (Figure 2b). During the experiments, all sensing modalities were recorded simultaneously with synchronized timestamps: the distance sensor at 20 Hz, the tactile sensor at 10 Hz, and the camera at 30 fps.

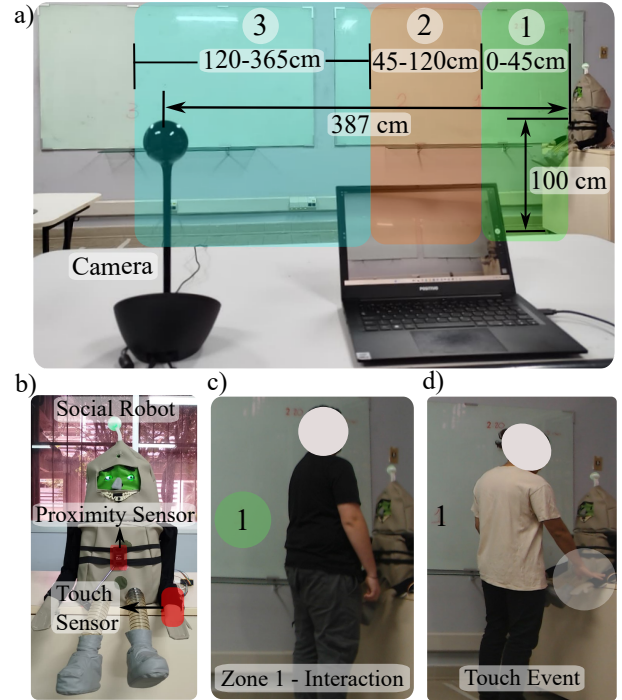


Fig. 2: Experimental setup for proximity and contact sensing. (a) Spatial configuration and interaction zones, (b) social robot with integrated sensing systems, (c) interaction scenario in Zone 1, and (d) Touch event.

Proximity: Proximity detection was evaluated when participants approached the robot under predefined conditions, following the spatial configuration and interaction zones shown in Figure 2a. Each trial started with the participant positioned in the public zone (>365 cm) for 5 s to establish a baseline. Participants were then instructed to walk toward the robot and stop at a target distance corresponding to one of the predefined social-distance zones: intimate (0–45 cm), social (45–120 cm), or public (120–365 cm). Each zone was maintained for 5 s, after which the participant moved sequentially through the zones—first public, then social, and finally intimate—and then reversed the sequence, returning outward while again holding each zone for 5 s. This sequence constituted one cycle, which was repeated five times.

Contact: Physical contact events were recorded through a

controlled protocol designed to generate consistent and repeatable interactions, as illustrated in Figure 2d. Each trial began with the participant standing in front of the robot at a neutral position with no contact for 2 s to establish a baseline. The participant is then asked to execute one contact on the robot’s hand, where the touch sensors are located underneath. Each contact was repeated across ten trials. After completing the interaction, the participant returned to the neutral position and waited 3–5 s before the next trial.

C. Data management

All sensor streams were logged with timestamps and stored in synchronized time-series format. Ground-truth labels for both contact events and proximity zone transitions were obtained through frame-by-frame video annotation. For each trial, annotators marked the onset and offset of physical contact as well as the moments when participants entered or exited each predefined distance zone by using floor markers. Using these annotations, event-level detection outcomes are obtained for each sensing modality. Each sample was classified as true positive (TP), false positive (FP), or false negative (FN) relative to the ground-truth event boundaries to obtain the accuracy and error rate for each system. Detection latency was quantified as the temporal difference between the annotated event onset and the corresponding detection time. Latency values are reported for individual sensing modalities. For the redundant systems, latency was not evaluated because the fusion output was obtained from synchronized offline streams rather than from a fully deployed real-time implementation.

IV. RESULTS

This section presents the experimental results obtained for proximity and physical contact detection, the performance of the singles and redundant sensing strategies, and quantitative results are reported using confusion matrices and standard performance metrics.

A. Approximation social areas

Figure 3a and b presents the confusion matrices obtained for proximity detection using the camera-based system and the distance sensor. In both cases, most predictions are concentrated along the main diagonal, indicating high classification performance across the evaluated proxemic zones. Table 1 summarizes the performance metrics obtained for proximity detection. The camera-based system achieved the highest overall accuracy (0.984), with high precision, recall, and F1-score values across all three zones. Precision and recall of 1.000 were observed in Zone 3, while a slightly lower recall value was ob-

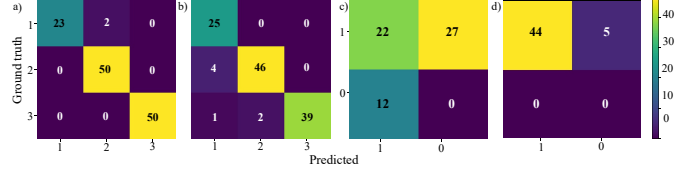


Fig. 3: Confusion matrices for proximity and touch detection: (a) camera-based proximity, (b) distance sensor system, (c) camera-based touch, and (d) tactile sensor system.

tained in Zone 1, which is expected near zone transition boundaries. These results are comparable to or exceed those reported in vision-based proxemics estimation studies under controlled indoor conditions [13].

The distance sensor achieved an overall accuracy of 0.940. Lower precision was observed in Zone 1 (0.833), whereas recall values remained high across the evaluated zones. Similar boundary-related errors have been reported in time-of-flight-based proximity sensing due to measurement noise and sensitivity to small distance variations [16]. The redundant sensing strategy achieved an accuracy of 0.960, with balanced precision, recall, and F1-score values across the three zones. F1-score values for the redundant system remained above 0.920 in all cases, indicating improved robustness compared to individual sensing modalities. Finally, detection latency was 0.30 s for the camera-based system and 0.19 s for the distance sensor, which is consistent with the expected processing overhead of vision-based approaches. Latency was not reported for the redundant proximity system because the fusion output was computed from offline synchronized streams.

Table 1: Performance metrics for proximity detection.

Method	Class	Precision	Recall	F1-score	Accuracy	Latency
Camera	Zone 1	1.000	0.920	0.958	0.984	0.30s
	Zone 2	0.962	1.000	0.980		
	Zone 3	1.000	1.000	1.000		
Sensor	Zone 1	0.833	1.000	0.909	0.940	0.19s
	Zone 2	0.958	0.920	0.939		
	Zone 3	1.000	0.929	0.963		
Redundant	Zone 1	0.920	0.920	0.920	0.960	-
	Zone 2	0.941	0.960	0.950		
	Zone 3	1.000	0.980	0.990		

B. Physical contact detection

Figure 3c and d show the confusion matrices obtained for physical contact detection using the camera and the tactile sensor. The camera-based system presents a higher number of missed detections, whereas the tactile sensor exhibits a higher concentration of correct classifications. Table 2 summarizes the performance metrics obtained for physical contact detection. The camera-based system achieved a precision of 0.65 and a

Table 2: Performance metrics for touch detection.

Metric	Camera	Sensor	Redundant
Precision	0.647	1.000	0.803
Recall	0.449	0.898	1.000
F1-score	0.530	0.946	0.891
Latency	0.41s	0.01s	-

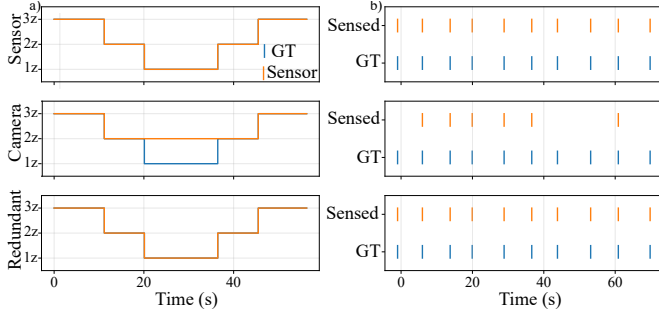


Fig. 4: Example signals illustrating the ground truth (GT) and sensed detection for the single sensors and redundant systems over time. a) proximity zone evolution and b) corresponding touch events.

recall of 0.45, resulting in an F1-score of 0.53. The tactile sensor achieved higher performance, with a precision of 1.00, a recall of 0.90, and an F1-score of 0.95. The redundant touch detection system achieved the highest recall value (1.00), while maintaining a precision of 0.80 and an F1-score of 0.89. Finally, detection latency was 0.41 s for the camera-based system and 0.01 s for the tactile sensor. Latency for the redundant touch system was not evaluated for the same reason, since fusion decisions were derived from offline synchronized sensor streams.

V. DISCUSSION

A. Approximation

The proximity detection results demonstrate that all evaluated systems are capable of identifying proxemic zones with high agreement relative to the ground-truth annotations, as shown in the example in Figure 4a. Among the individual sensing modalities, the camera-based approach achieved the highest overall accuracy (0.984), reflecting its ability to continuously monitor the interaction distance. Similar performance trends for vision-based proximity estimation have been reported in previous HRI studies conducted under controlled environmental conditions [17], [18].

The distance sensor achieved a slightly lower accuracy (0.940), with most classification errors occurring near the boundaries between adjacent proxemic zones. This behavior is consistent with characteristics commonly reported for ToF sensing technologies, where small measurement fluctuations can lead to ambiguity during zone transitions [19]. Despite

this limitation, ToF-based sensors remain widely used in HRI due to their compact size, low computational requirements, and direct distance measurement capabilities [19]. The redundant sensing strategy fulfilled its intended role of enhancing system reliability by stabilizing proximity estimation when discrepancies between sensing modalities occurred. This behavior aligns with prior work on redundant and multimodal perception frameworks, which emphasize robustness and fault tolerance as primary objectives of sensor fusion in HRI [18], [20].

B. Physical human–robot interactions

For pHRI detection, substantial performance differences were observed between the camera-based system and the tactile sensor. This limitation is commonly associated with vision-based contact detection methods and is typically attributed to occlusions, viewpoint constraints, and reliance on visual overlap cues between detected objects [17]. In contrast to highly integrated tactile sensors such as F-VIS [21], which focus on precise multimodal force reconstruction at the sensor level, the present work addresses contact and proximity detection from a system-level HRI perspective. F-VIS demonstrates outstanding performance in terms of force resolution, depth reconstruction, and multimodal fusion by combining FBG-based sensing with vision-based tactile imaging, enabling fine manipulation. However, such approaches rely on specialized hardware, complex calibration procedures, and tightly coupled sensing architectures, which may limit their scalability and deployment in social robots operating in unstructured, human-centered environments. Rather than aiming for high-precision force estimation, the proposed system focuses on reliable detection of contact events and proxemic zones, achieving high recall for physical contact and high accuracy for proximity detection even in the presence of noise or partial sensor failure. Argall *et al.* reported similar results in pHRI with the single touch sensor, where tactile and pressure-based sensors provide reliable and low-latency detection of physical contact events [22], [23].

C. Limitations

It is important to acknowledge the limitations of this study. Although the experiments simulated real-time sensor acquisition, they do not fully capture real-world deployment constraints such as computational load, lighting conditions, and occlusions, particularly for vision-based methods that rely on deep learning and are sensitive to environmental factors. Second, proximity and contact events were analyzed using temporal intervals derived from ground-truth annotations rather than exact event timestamps. While this approach ensures consistency across sensing modalities, it may obscure fine-grained temporal discrepancies between systems, especially during rapid transitions between interaction states. Finally, the experi-

mental setup was conducted under controlled conditions with a fixed camera viewpoint and predefined interaction patterns. Future work will focus on improving key algorithmic components of the system. The YOLOv8-based vision pipeline for proximity estimation will be optimized for real-time inference and increased robustness to illumination changes and occlusions. In addition, the camera-based touch detection and the current fusion strategies (logical OR and minimum-delta rules) will be refined to improve reliability in real-world HRI scenarios.

VI. CONCLUSION

This work presented a redundant monitoring system for proximity and physical contact detection in HRI scenarios, combining camera-based and sensor-based approaches. For proximity estimation, both the camera-based and distance sensor-based systems demonstrated high accuracy in identifying proxemic zones. The redundant sensing strategy provided a balanced solution, achieving an accuracy of 0.960 by increasing robustness against transient errors and sensor-specific limitations. In physical contact detection, the redundant touch detection system, implemented using a logical OR fusion strategy, achieved high recall and precision, ensuring that all contact events were detected. This behavior is particularly relevant in HRI contexts, where failing to detect physical contact can compromise user safety or user perception.

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